## HEMAL SAI MANYAM

**Prospective PhD Researcher | Cybersecurity | Adversarial Security | Secure Systems**

Munich, Germany

**Email:** hemalsaimanyam5@gmail.com | **Phone:** +49 155 6079 6480

**LinkedIn:** linkedin.com/in/hemalsai


## RESEARCH PROFILE

M.Sc. Cybersecurity graduate from HDBW Munich with research experience in adversarial machine learning, AI-assisted intrusion detection, digital forensics, and experimental security evaluation. My master's research investigated how adversarial behaviour affects the reliability of security detection and forensic evidence, supported by a self-developed, segmented cybersecurity laboratory.

Alongside academic research, I gained industry experience at Allianz Technology and Reliance Jio, working with security architectures, identity and access management, enterprise infrastructure, and security monitoring.

## EDUCATION

**Master of Science (M.Sc.) - Cybersecurity**

HDBW University of Applied Sciences, Munich, Germany | 2026

- Master's Thesis: Adversarial Attacks on AI-Powered Cyber Defense Systems and Their Impact on Digital Forensics
- Research areas: Adversarial Machine Learning, Intrusion Detection, Digital Forensics, Security Monitoring, and Experimental Security Evaluation.
- Relevant coursework: Network Security, Cryptography, Artificial Intelligence, Digital Forensics, Cloud Security, and Secure Software Development.

## MASTER'S THESIS & RESEARCH EXPERIENCE

**Adversarial Attacks on AI-Powered Cyber Defense Systems and Their Impact on Digital Forensics**

Master's Research Project | HDBW Munich | 2026

**Research Objective:** Investigated how adversarial techniques affect AI-assisted security detection and the completeness, consistency, and reliability of digital forensic evidence.

- Designed and implemented a segmented cybersecurity testbed integrating attack-generation systems, network security monitoring, application services, and centralized security analytics.
- Developed Python-based security detection pipelines using Logistic Regression, Random Forest, XGBoost, Isolation Forest, and TF-IDF-based analysis.
- Simulated adversarial scenarios including network reconnaissance, low-rate scanning, authentication attacks, SQL injection, and application-level manipulation.
- Integrated Wazuh SIEM, Suricata IDS, Zeek, pfSense, and application telemetry to examine security events across multiple detection layers.
- Evaluated the effects of adversarial behaviour on detection performance, evidence completeness, temporal coherence, and cross-layer correlation.
- Investigated how network segmentation, sensor placement, and logging limitations influence the interpretation of security incidents.
- Conducted experimental troubleshooting involving Linux packet filtering, reverse-path filtering, routing, NAT, and network connectivity.

**Research Contribution:** Demonstrated how adversarial behaviour and monitoring limitations can create gaps between observed attack activity, AI-generated alerts, and the evidence available for subsequent forensic investigation.

## PROFESSIONAL EXPERIENCE

**Allianz Technology SE - Munich, Germany**

Working Student - Information Security | March 2025 - March 2026

- Supported security architecture reviews, threat modelling, and technical security risk assessments across enterprise systems.
- Analysed identity and privileged-access architectures involving Microsoft Entra ID/PIM, CyberArk, MFA, RBAC, and Zero Trust principles.
- Evaluated AI/LLM security risks, including prompt injection, data exposure, and model manipulation.
- Collaborated with technical stakeholders to identify security weaknesses and recommend appropriate controls.
- Developed Power Automate and SharePoint workflows to support security reporting and operational processes.
- Contributed to security awareness and technical knowledge-sharing initiatives.

**Reliance Jio Platforms Limited - Mumbai, India**

Deputy Manager - Information Security | March 2023 - August 2024

- Worked on enterprise authentication and authorization infrastructure using OAuth 2.0, OpenID Connect, SAML, LDAP, RADIUS, and MFA.
- Supported security operations across Linux systems, enterprise applications, and containerized environments.
- Investigated technical failures involving authentication, network connectivity, certificates, and application infrastructure.
- Used Splunk, ELK, Grafana, and Prometheus for monitoring, log analysis, and troubleshooting.
- Worked with Kubernetes, OpenShift, RHEL, Oracle WebLogic, and TLS/SSL configurations.

## TECHNICAL & RESEARCH SKILLS

**Security Research:** Experimental Security Evaluation • Adversarial Machine Learning • Threat Modelling • Intrusion Detection • Digital Forensics • Security Testing

**Programming & Data Analysis:** Python • Bash/Shell • Java • SQL • pandas • scikit-learn • REST APIs

**Machine Learning:** Logistic Regression • Random Forest • XGBoost • Isolation Forest • TF-IDF • Anomaly Detection • Classification

**Security Tools:** Wazuh • Suricata • Zeek • Wireshark • Nmap • Nessus • Burp Suite • Kali Linux • Splunk • ELK

**Systems & Networking:** Linux • Windows • VMware • pfSense • TCP/IP • Routing • NAT • Firewalls • Network Segmentation • Kubernetes • Docker • OpenShift

**Security Concepts:** Cryptography Fundamentals • Authentication Protocols • Access Control • Secure Systems • Network Security • Vulnerability Assessment

## RESEARCH INTERESTS

- **Implementation Security:** Understanding how security weaknesses emerge from system implementations, configurations, and operational behaviour.
- **Adversarial Resilience:** Evaluating how systems and security mechanisms respond to intentional manipulation.
- **Experimental Security Analysis:** Designing reproducible testbeds and controlled experiments to investigate security vulnerabilities.
- **Secure Systems:** Studying trust boundaries, attack surfaces, and security mechanisms across complex computing environments.
- **Trustworthy AI for Cybersecurity:** Improving the reliability, robustness, and interpretability of AI-assisted security decisions.